SEQUENCES

SERIES

GENERATING FUNCTIONS

SUMMABILITY

PROBABILITY

APPROXIMABILITY

I:

Everything You Knew or Need to Know About Sequences and Series

Dr. Gerald Higdon

Convergence of a Sequence

Definition I.1: A sequence $\{a_n\}$ is **convergent** with limit L, written $\lim_{n\to\infty} a_n = L$, if

for every $\varepsilon > 0$ there is an integer n_0 such that if $n \geq n_0$ then $|a_n - L| < \varepsilon$.

$$\forall\, \varepsilon > 0 \quad \exists\, n_0 \in \mathbb{N} \ni n \geq n_0 \implies |a_n - L| < \varepsilon.$$

Two Useful Limit Theorems:

Theorem I.2: If $\lim_{x\to+\infty} f(x) = L$ and $f(n) = a_n$, then $\lim_{n\to\infty} a_n = L$.

Theorem I.3: If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Degrees of Zero

Let $p, q > 0$ and $0 < r < 1$.

Then let

$$\ell_n = \frac{1}{(\ln n)^q}, \qquad p_n = \frac{1}{n^p}, \qquad g_n = r^n, \qquad f_n = \frac{1}{n!}, \qquad d_n = \frac{1}{n^n}.$$

Definition I.4: Assume $a_n > 0$, $b_n > 0$, $\lim_{n\to\infty} a_n = 0$, $\lim_{n\to\infty} b_n = 0$.

We say $\{a_n\}$ is **much less than** $\{b_n\}$, written $a_n \ll b_n$, if

$$\lim_{n\to\infty} \frac{a_n}{b_n} = 0.$$

Obviously, if $a_n \ll b_n$ and $b_n \ll c_n$, then $a_n \ll c_n$.

Degrees of Zero

Theorem I.5:

With $\{\ell_n\}, \{p_n\}, \{g_n\}, \{f_n\}, \{d_n\}$ as in Definition I.4,

$$d_n \ll f_n \ll g_n \ll p_n \ll \ell_n,$$

that is,

$$\frac{1}{n^n} \ll \frac{1}{n!} \ll r^n \ll \frac{1}{n^p} \ll \frac{1}{(\ln n)^q}.$$

Degrees of Zero

Example I.6: $g_n \ll p_n$

Let $p_n = \dfrac{1}{n^{100}}$ and $0 < r < 1$. With $b = \frac{1}{r} > 1$:

$$\lim_{x\to+\infty} \frac{r^x}{1/x^{100}} = \lim_{x\to+\infty} \frac{x^{100}}{b^x} \qquad \left(\frac{\infty}{\infty}\right)$$

$$\xrightarrow{\text{L.H.}} \frac{100\,x^{99}}{b^x \ln b} \quad \left(\frac{\infty}{\infty}\right) \quad \xrightarrow{\text{L.H.}} \cdots \xrightarrow{\text{L.H.}} \frac{100!}{b^x (\ln b)^{100}} \xrightarrow[x\to+\infty]{} 0.$$

Example I.7: $d_n \ll f_n$

$$\lim_{n\to\infty} \frac{d_n}{f_n} = \lim_{n\to\infty} \frac{1/n^n}{1/n!} = \lim_{n\to\infty} \frac{n!}{n^n}$$

$$= \lim_{n\to\infty} \frac{n\,(n-1)\cdots 2\cdot 1}{n\cdot n \cdots n\cdot n}$$

$$\leq \lim_{n\to\infty} \frac{1}{n} = 0\ .$$

Powers of r and Monotone Sequences

Remark I.8: The sequence $\{r^n\}$ is convergent iff $-1 < r \leq 1$, and

$$\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1, \\ 1 & \text{if } r = 1. \end{cases}$$

Theorem I.9: Every bounded monotonic sequence converges.

From Sequences to Series

Definition I.10: For a series $\sum_{n=1}^{\infty} a_n$, let $\{S_n\}$ be its **sequence of partial sums**, where

$$S_n = \sum_{k=1}^{n} a_k.$$

If $\{S_n\}$ converges with limit $S \in \mathbb{R}$, the series is convergent with sum S.

If $\{S_n\}$ is divergent, the series $\sum_{n=1}^{\infty} a_n$ is divergent.

Geometric Series

$$\sum_{n=1}^{\infty} ar^{n-1} = \sum_{n=0}^{\infty} ar^{n} \quad \text{is convergent} \iff |r| < 1,$$

in which case

$$\sum_{n=0}^{\infty} ar^{n} = a\left(1 + r + r^2 + \cdots\right)$$

$$= \frac{a}{1-r}.$$

Convergence

Theorem I.11: If $\sum_{n=1}^{\infty} a_n$ is convergent, then $\lim_{n\to\infty} a_n = 0$.

Corollary I.12: If $\lim_{n\to\infty} a_n \neq 0$ or $\lim_{n\to\infty} a_n$ does not exist, then $\sum_{n=1}^{\infty} a_n$ diverges.

Theorem I.13: If $\sum a_n$, $\sum b_n$ converge, then $\sum c\, a_n$ and $\sum(a_n \pm b_n)$ also converge, and:

$$\sum c\, a_n = c \sum a_n \quad \text{and} \quad \sum(a_n \pm b_n) = \sum a_n \pm \sum b_n.$$

The Integral Test

Theorem I.14 (Integral Test):

Let f be continuous, positive, decreasing on $[1, +\infty)$ with $a_n = f(n)$.

Then if

$$\int_1^{+\infty} f(x)\,dx \text{ converges, so does } \sum_{n=1}^{\infty} a_n,$$

and if

$$\int_1^{+\infty} f(x)\,dx \text{ diverges, so does } \sum_{n=1}^{\infty} a_n.$$

The Comparison Test

Corollary I.15: $\sum_{n=1}^{\infty} \frac{1}{n^p}$ is convergent if $p > 1$ and divergent if $p \leq 1$.

Theorem I.16 (Comparison Test): Assume $a_n > 0$, $b_n > 0$.

If $\sum b_n$ converges and $a_n \leq b_n$ for all n, then $\sum a_n$ converges.

If $\sum b_n$ diverges and $a_n \geq b_n$ for all n, then $\sum a_n$ diverges.

THE "WALL" OF CONVERGENCE FOR SERIES:

$$\underbrace{\frac{1}{n^n},\ \frac{1}{n!},\ \ldots,\ r^n,\ \frac{1}{n^p}\,(p>1)}_{\textbf{converge}} \quad\Big|\quad \underbrace{\frac{1}{n^p}\,(p\leq 1),\ \frac{1}{(\ln n)^q}}_{\textbf{diverge}}$$

Limit Comparison Test

Theorem I.17 (Limit Comparison Test):

Assume $a_n > 0$, $b_n > 0$. If $\lim_{n\to\infty} \frac{a_n}{b_n} = c > 0$, then both series converge or both diverge.

If $\lim_{n\to\infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, so does $\sum a_n$.

If $\lim_{n\to\infty} \frac{a_n}{b_n} = +\infty$ and $\sum b_n$ diverges, so does $\sum a_n$.

Alternating Series

Definition I.18: Let $b_n > 0$. An alternating series is a series of the form

$$\sum_{n=1}^{\infty}(-1)^{n-1}b_n.$$

Theorem I.19 (Alternating Series Test): If $b_{n+1} < b_n$ for all n and $\lim_{n\to\infty} b_n = 0$, then $\sum_{n=1}^{\infty}(-1)^{n-1}b_n$ converges.

Definition I.20: A series $\sum a_n$ is absolutely convergent if $\sum |a_n|$ converges.

Theorem I.21: If $\sum a_n$ is absolutely convergent, then $\sum a_n$ is convergent.

The Ratio Test

Theorem I.22 (Ratio Test): Consider a series $\sum a_n$.

1) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L < 1$, then $\sum a_n$ is absolutely convergent.

2) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = L > 1$ or $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = +\infty$, then $\sum a_n$ diverges.

3) If $\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = 1$, there is no conclusion.

The Root Test

Theorem I.23 (Root Test): Consider a series $\sum a_n$.

1) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L < 1$, then $\sum a_n$ is absolutely convergent.

2) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = L > 1$ or $= +\infty$, then $\sum a_n$ is divergent.

3) If $\lim_{n\to\infty} \sqrt[n]{|a_n|} = 1$, there is no conclusion.

Power Series

Definition I.24: A power series is a series of the form

$$\sum_{n=0}^{\infty} a_n(x-c)^n = a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots \ .$$

The a_n's are the coefficients and c is called the center.

Theorem I.25: Given $\sum_{n=0}^{\infty} a_n(x-c)^n$, there are three possibilities:

1) the series converges only for $x = c$;

2) the series converges absolutely for all x;

3) there is an $R > 0$ such that the series converges absolutely for $|x-c| < R$ and diverges for $|x-c| > R$.

Intervals of Convergence

Remark I.26:

1) In case (3) of **Theorem I.25**, the series may or may not converge for $|x - c| = R$, i.e. $x = c \pm R$.

2) Referring to the cases of **Theorem I.25**, the intervals of convergence are as follows:

Case (1): $\{c\}$.

Case (2): $\mathbb{R} = (-\infty, +\infty)$.

Case (3): one of $(c - R, c + R)$, $[c - R, c + R)$, $(c - R, c + R]$, or $[c - R, c + R]$.

3) The R in case (3) of Theorem I.25 is called the radius of convergence. In case (1) $R = 0$, while $R = +\infty$ in case (2) of **Theorem I.25**.

Taylor Series

Theorem I.27: Assume $\sum_{n=0}^{\infty} a_n(x-c)^n$ has $R > 0$ (possibly $R = +\infty$).

Let $f(x) = \sum_{n=0}^{\infty} a_n(x-c)^n$. Then

$$a_n = \frac{f^{(n)}(c)}{n!}, \text{ and the series}$$

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(c)}{n!}(x-c)^n \text{ is called a Taylor Series.}$$

If $c = 0$, the series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n$ is called a Maclaurin series.

Important Maclaurin Series

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n = 1 + x + x^2 + \cdots, \qquad -1 < x < 1.$$

$$\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - \cdots, \qquad -1 < x < 1.$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \qquad -\infty < x < +\infty.$$

Important Maclaurin Series

$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots.$$

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots.$$

$$\cosh x = \sum_{n=0}^{\infty} \frac{x^{2n}}{(2n)!} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \cdots.$$

$$\sinh x = \sum_{n=0}^{\infty} \frac{x^{2n+1}}{(2n+1)!} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \cdots.$$

All valid for $-\infty < x < +\infty$.

Preview

Formal Power Series:

$$A(x) = \sum_{n=0}^{\infty} a_n x^n \qquad \longleftrightarrow \qquad \{a_0, a_1, \ldots\}$$

Definitions:

1) **Addition / Subtraction:**

$$A(x) \pm B(x) = \sum_{n=0}^{\infty} a_n x^n \pm \sum_{n=0}^{\infty} b_n x^n = \sum_{n=0}^{\infty} (a_n \pm b_n)\, x^n.$$

2 **Multiplication**

$$A(x) \cdot B(x) = \sum_{n=0}^{\infty} c_n x^n \qquad \text{where} \qquad c_n = \sum_{k=0}^{n} a_k\, b_{n-k}.$$

Application / Example

Let $a_0 = 1,\ a_1 = -1,\ a_n = 0$ for $n \geq 2$.

Let $b_n = 1$ for all n.

$$c_n = \sum_{k=0}^{n} a_k\, b_{n-k} = \sum_{k=0}^{n} a_k.$$

$$c_0 = a_0 = 1, \quad c_1 = a_0 + a_1 = 0, \quad c_n = a_0 + a_1 + \sum_{k=2}^{n} a_k = 0.$$

Conclusion:

$$(1 - x)\left(1 + x + x^2 + \cdots\right) = 1.$$

Generating Function

Definition: Let $\{a_0, a_1, a_2, \ldots\}$ be a sequence of numbers. Its generating function (generating series) is

$$a_0 + a_1 s + a_2 s^2 + \cdots = \sum_{n=0}^{\infty} a_n s^n.$$

Definition: The symbol $f \overset{\text{ops}}{\longleftrightarrow} \{a_n\}_{n=0}^{\infty}$ means f is the ordinary (formal) power-series generating function for the sequence $\{a_n\}_{n=0}^{\infty}$. That is

$$f = \sum_{n=0}^{\infty} a_n x^n.$$

Definition: A generating function is the representation of an infinite sequence of numbers as the coefficients of a formal power series.